BRSM Report 1

Validation of a Gamified Selective Attention Task Against a Laboratory Standard: A Mixed-Design Behavioral Study

Team 4rp3n4s
(Ananya Kasavajhala, Rohit Jeswanth, Vishak Kashyap K)

Abstract

Gamification is increasingly being used in cognitive assessment and training, but its validity compared to standard laboratory tasks must be empirically established. This study evaluates a newly developed *Selective Attention Game* by comparing it with a conventional lab-based attention task based on Feature Integration Theory (Treisman & Gelade, 1980). A 2 x 2 mixed factorial design was used with Target Load (Single vs Multiple) as a between-subjects factor and Modality (Game vs Lab) as a within-subjects factor. Performance was measured using Reaction Time (RT) and Accuracy. Descriptive statistics, correlation analysis, mixed ANOVA, t-tests, regression and reliability analysis were conducted. Results showed significant main effects of Target Load and Modality, along with a strong interaction effect for reaction time. Concurrent validity between game and lab RT was not statistically significant. Accuracy patterns differed substantially across conditions. Reliability of level-wise RT scores was low. The findings provide an initial statistical understanding of performance differences between gamified and laboratory attention tasks.

Introduction

Selective attention is the cognitive process through which relevant stimuli are prioritised while irrelevant information is filtered out. One influential theory of attention is the Feature Integration Theory (FIT) proposed by Treisman and Gelade in 1980. FIT distinguishes between:

Feature Search: Targets defined by a single visual feature (like colour) are processed in parallel and detected quickly with minimal errors.

Conjunction Search: Targets defined by multiple features (like color + shape) require serial attentional binding. They are slower and more error-prone.

In the present experiment, these two types of search were implemented as:

Condition	**Type of Search**	**Expected Performance**
Single Target	Feature Search	Fast, almost perfect accuracy
Multiple Target	Conjunction Search	Slower, prone to error

With the rise of mobile cognitive tools, gamified interfaces are being developed for attention assessment. Systematic reviews indicate that gamification can influence engagement and performance in cognitive tasks (Lumsden et al., 2016). Recent reviews also highlight the growing use of digital games specifically for attention assessment and the need for careful validation of such tools (Wiley et al., 2021). However, their performance must be validated against established laboratory tasks. Previous studies evaluating the validity of game-based cognitive assessments have reported mixed findings regarding their comparability with traditional measures (Aneni et al., 2023).

Objective

To validate a newly developed Selective Attention Game by comparing performance with a standard lab task.

Research Questions

RQ1. Concurrent Validity: Is there a significant positive correlation between performance (RT/Accuracy) in the Game and the Lab Task?

RQ2. Effect of Target Load: Is performance significantly worse in the Multiple Target condition compared to the Single Target condition?

RQ3. Effect of Modality: Does the gamified interface significantly alter performance compared to the standard lab interface?

RQ4. Effect of Game Level: What is the effect of level in the games? (performance change across game levels)

Methods

Design

The following 2 x 2 Mixed Factorial design was used:

Factor	**Type**	**Levels**
Target Load	Between-subjects	Single, Multiple
Modality	Within-subjects	Lab, Game

Participants

Total sample: N = 37
Participants were randomly assigned to target load groups:
- Single Target group: n = 21
- Multiple Target group: n = 16

All participants completed both modalities. Task order was counterbalanced, with half the participants completing the game first and the remaining completing the lab task first.

Measures

Variable	**Description**
Reaction Time (RT)	Response latency in milliseconds
Accuracy	Percentage of correct responses

Data Processing

Lab Task

- RT: First click latency
- Single-target accuracy: Correct if target clicked
- Multiple-target accuracy: Correct if no error feedback is triggered

Game Task

- RT: Initial Response time (in ms)
- Accuracy: Success rate (in %)

Statistical Analyses

The following were conducted:

1. Descriptive statistics (Mean, SD)
2. Pearson Correlation (Concurrent validity)
3. Mixed 2 x 2 ANOVA
4. Paired and independent t–tests
5. Linear Regression (for level effect)
6. Reliability analysis (Cronbach's α, Split-half)

Results

Descriptive Statistics

Reaction Time (ms)

Target Load	**Lab (M ± SD)**	**Game (M ± SD)**
Single	1565.4 ± 236.5	2957.6 ± 520.8
Multi	1605.0 ± 273.6	1856.5 ± 507.3

Accuracy (%)

Target Load	Lab (M ± SD)	Game (M ± SD)
Single	100.00 ± 0.00	95.79 ± 5.56
Multi	40.83 ± 21.06	96.37 ± 2.53

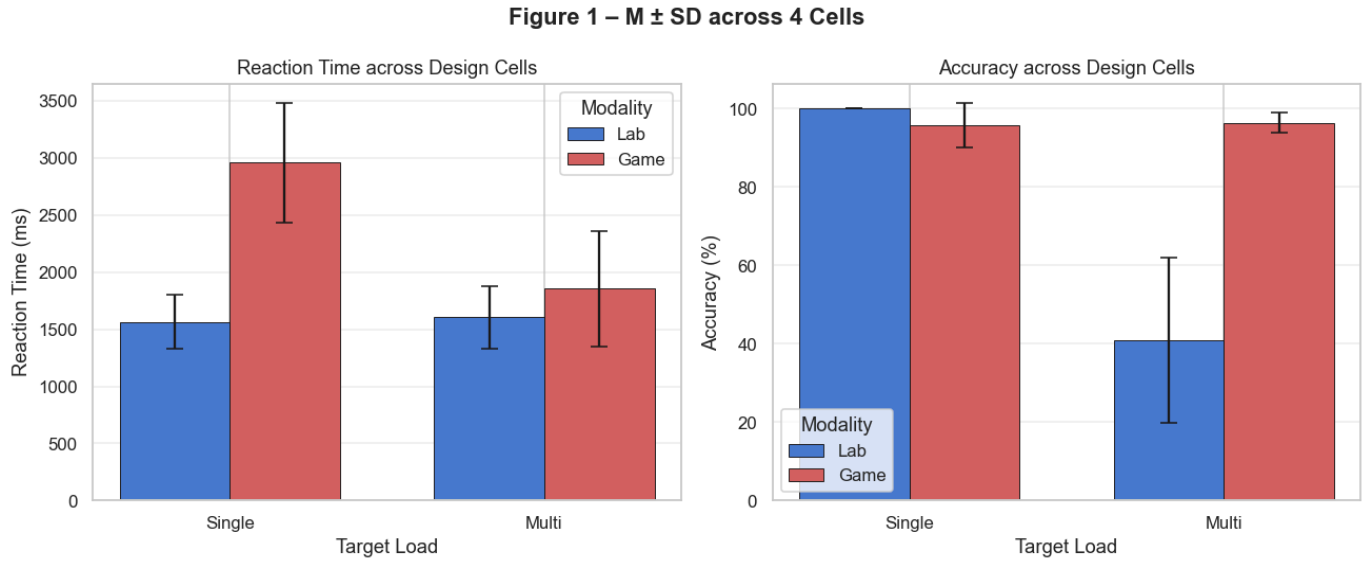

Figure 1. Mean reaction time and accuracy for each target load and modality (error bars show ±1 SD).

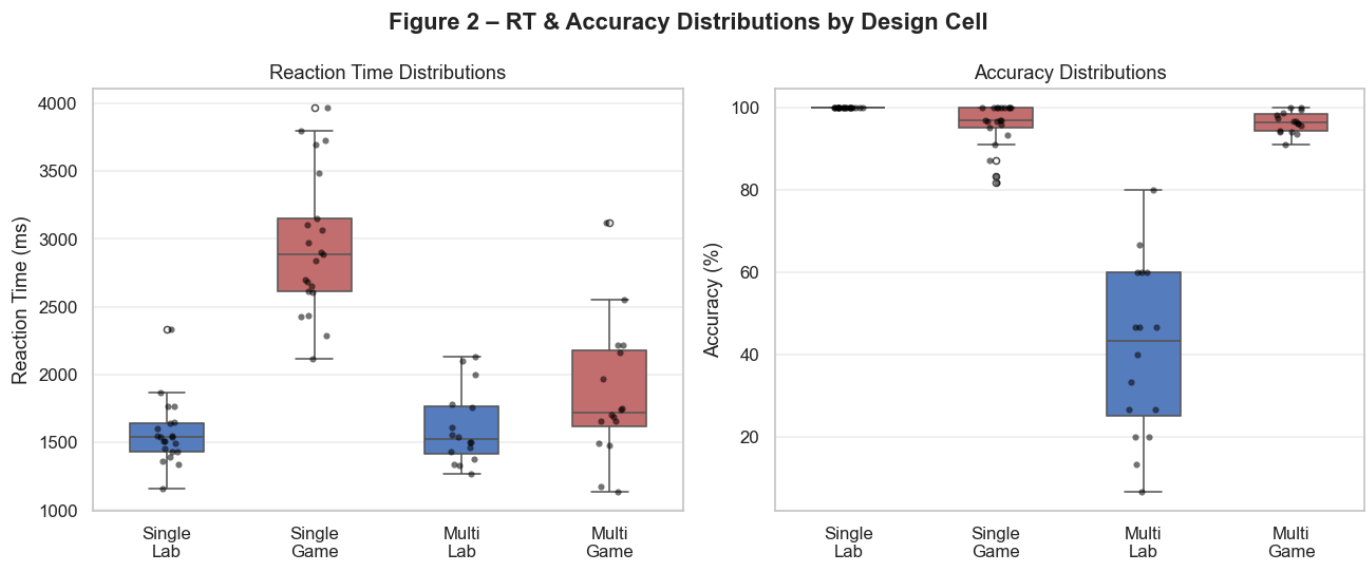

Figure 2. Distribution of reaction time and accuracy across design cells. Each point represents one participant.

RQ1: Concurrent Validity

Correlation between game and lab RT:

Group	Pearson r	p
Single	0.288	0.2055
Multi	0.101	0.7096

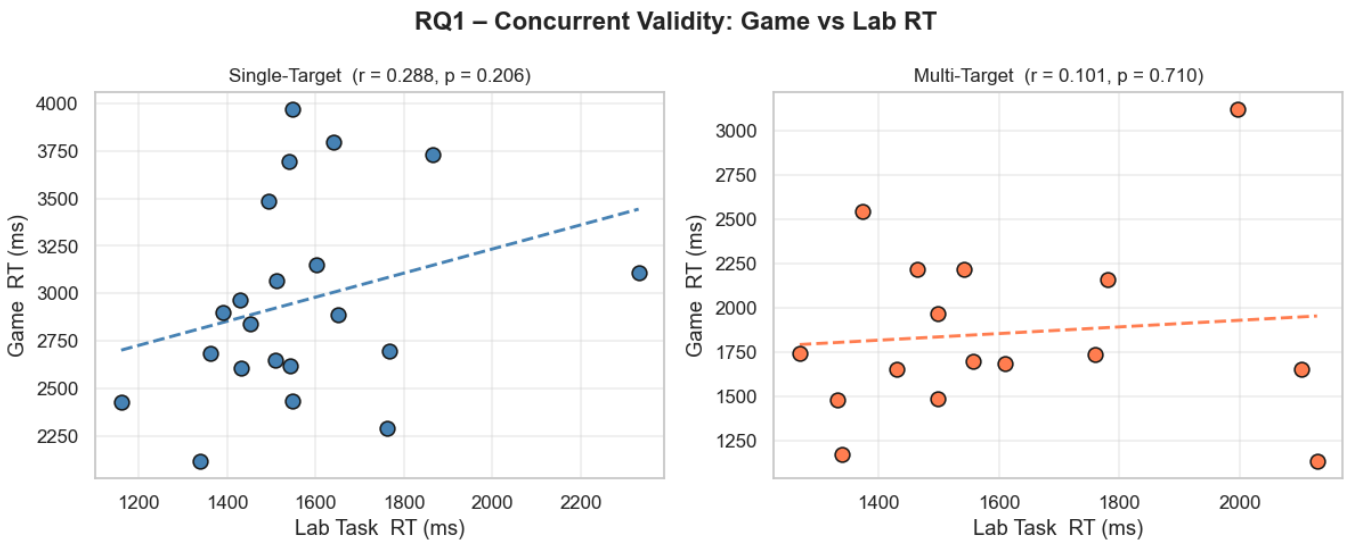

Figure 3. Scatterplots showing correlation between laboratory and game reaction times for single and multi target groups.

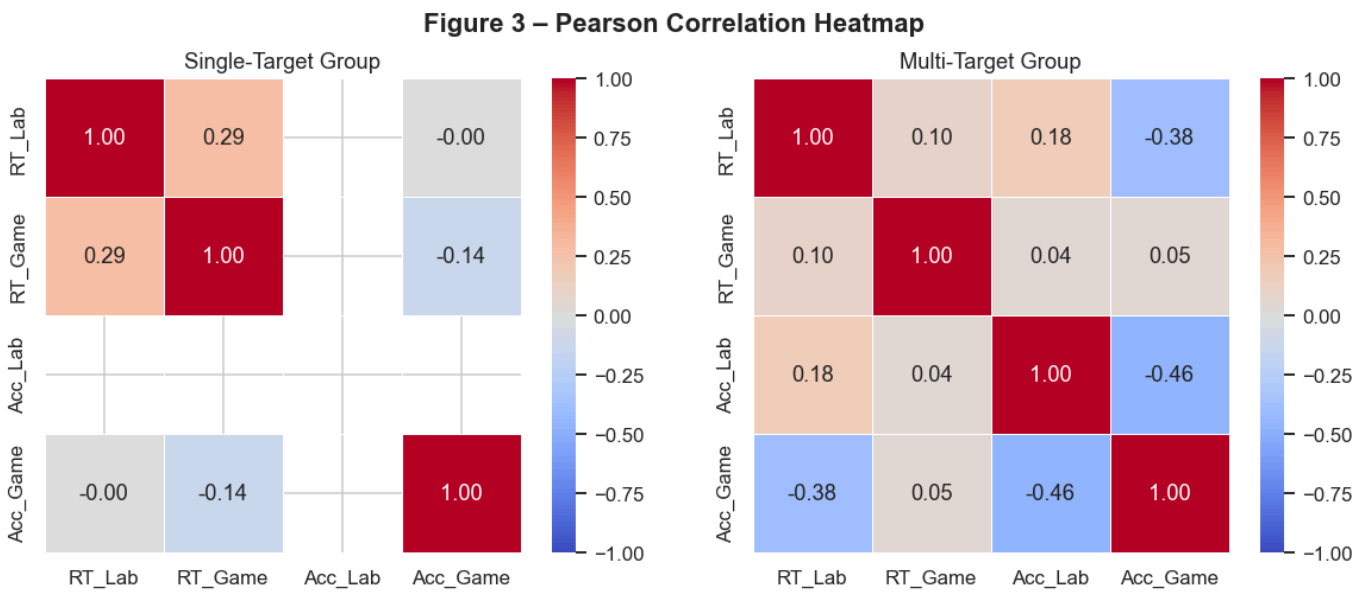

Figure 4. Pearson correlation matrices for performance metrics within single and multi target groups.

No statistically significant correlations were observed.

RQ2 & RQ3: Mixed ANOVA

Reaction Time

Source	**F(1, 35)**	**p**	**η^2_p**
Target Load	26.8	<0.001	0.43
Modality	108.0	<0.001	0.76
Interaction	42.7	<0.001	0.55

Accuracy

Source	**F(1, 35)**	**p**	**η^2_p**
Target Load	164.4	<0.001	0.82
Modality	74.7	<0.001	0.68
Interaction	140.0	<0.001	0.80

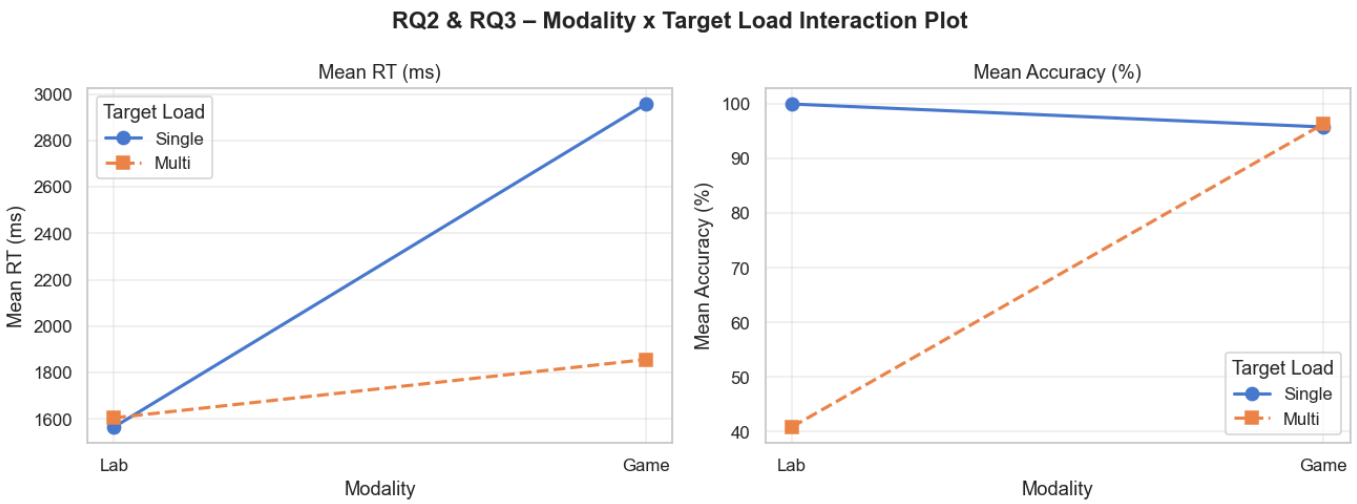

Figure 5. Interaction between target load and modality for reaction time and accuracy.

All main effects were statistically significant.

Pairwise Comparisons

Paired t-tests (Game vs Lab)

Group	DV	t(df)	p	Cohen's d
Single	RT	-12.603(20)	<0.001	-2.750
Multi	RT	-1.823(15)	0.088	-0.456
Single	Accuracy	3.469(20)	0.002	0.757
Multi	Accuracy	-9.943(15)	<0.001	-2.486

Independent t-tests (Single vs Multi)

Modality	DV	t	p	Cohen's d
Lab	RT	-0.462	0.647	-0.157
Game	RT	6.466	<0.001	2.138
Lab	Accuracy	11.235	<0.001	4.291
Game	Accuracy	-0.427	0.673	-0.129

RQ4: Effect of Level

Linear Regression (RT vs Level)

Group	β(ms/level)	r	p
Single	+261.25	0.445	<0.001
Multi	-25.11	-25.11	0.396

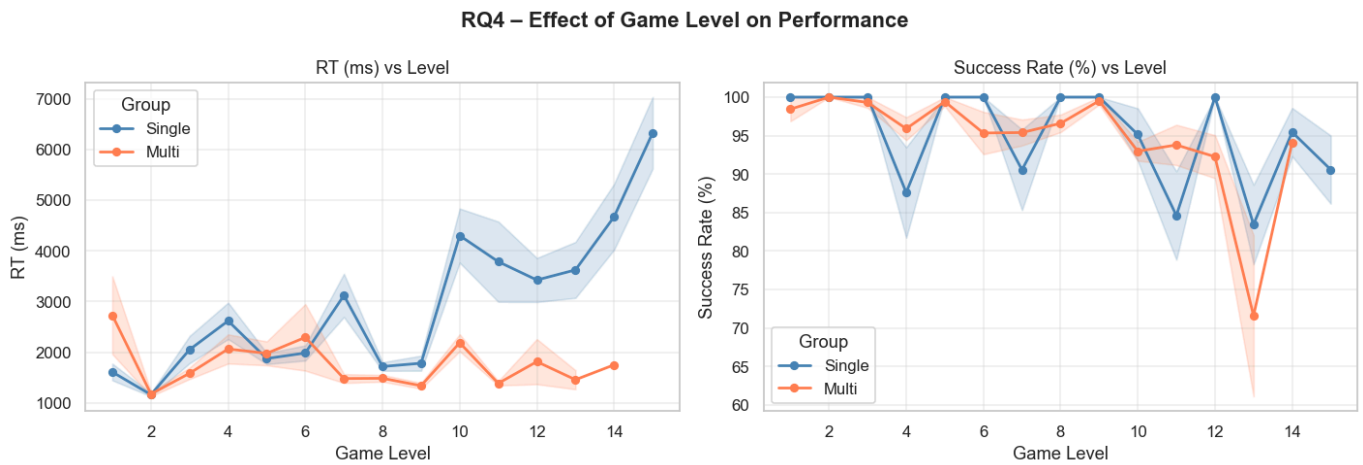

Figure 6. Reaction time and success rate across game levels for Single and Multi target groups.

RT increased significantly across levels only in the Single Target group.

ANOVA across levels 1-10 (pooled):

$F = 6.263$, $p < 0.001$

Reliability

Group	**Cronbach's α**	**Splithalf (SB-corrected)**
Single	0.138 (poor)	-0.436
Multi	0.476 (poor)	0.601

Internal consistency of level-wise RT was low.

Interpretation of Results

- There was a substantial difference across modality and task complexity. Reaction times were significantly slower in the game interface, particularly for the Single target group.
- Participants showed almost perfect accuracy in the lab based feature search task, as expected based on FIT.
- There was no strong correlation between modalities, suggesting that the concurrent validity of RT measures is weak. Similar concerns about the strength of validity evidence in game-based cognitive assessments have been noted in prior reviews (Aneni et al., 2023).
- Interaction effects were significant indicating that modality influence on performance differs depending on task load.
- Level-wise analysis shows increase in RT with difficulty progression in the Single Target group, but not in the Multi Target group.
- Reliability analysis shows low internal consistency of RT across levels. This suggests task difficulty or participant responses varied across levels

Conclusion

The gamified attention task shows statistically distinct performance characteristics compared to the standard lab task. While significant effects of modality and target load were observed, the concurrent validity of reaction time measures remains uncertain. Further improvement in measurement consistency may be needed.

Next Steps

- Practice Effects: Use linear regression to see if RT decreases as players get familiar with the game. Significant observations can be examined using regression diagnostics (Cook, 1977)

- Outlier Control: Apply Winsorization and non-parametric tests (Wilcoxon signed-rank) to handle extreme data spreads if normality assumptions remain violated (Siegel, 1956; Wilcoxon, 1945).
- Combined Predictors: Use Multiple Regression to see if Game RT and Accuracy together can predict Lab results.
- Error Breakdown: Use Chi-Square (χ^2) tests to see if complex tasks lead to more False Alarms vs. Misses.
- Bayesian Strength: Calculate the Bayes Factor to quantify the weight of evidence for differences between Lab and Game modalities, providing a quantitative measure of evidence beyond p-values (Rouder et al., 2009).

References

Aneni, K., Gomati de la Vega, I., Jiao, M. C. F., et al. (2023). Evaluating the validity of game-based assessments measuring cognitive function: A systematic review and meta-analysis. *Progress in Brain Research.* https://doi.org/10.1016/bs.pbr.2023.02.002

Cook, R. D. (1977). Detection of influential observation in linear regression. *Technometrics, 19*(1), 15–18. https://doi.org/10.2307/1268249

Lumsden, J., Edwards, E. A., Lawrence, N. S., Coyle, D., & Munafò, M. R. (2016). Gamification of cognitive assessment and cognitive training: A systematic review of applications and efficacy. *JMIR Serious Games, 4*(2), e11. https://doi.org/10.2196/games.5888

Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., & Iverson, G. (2009). Bayesian t tests for accepting and rejecting the null hypothesis. *Psychonomic Bulletin & Review, 16*(2), 225–237. https://doi.org/10.3758/PBR.16.2.225

Siegel, S. (1956). *Nonparametric statistics for the behavioral sciences.* McGraw-Hill.

Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology, 12*(1), 97–136. https://doi.org/10.1016/0010-0285(80)90005-5

Wiley, K., Robinson, R. L., & Mandryk, R. L. (2021). The making and evaluation of digital games used for the assessment of attention: A systematic review. *JMIR Serious Games, 9*(3), e26449. https://doi.org/10.2196/26449

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin, 1*(6), 80–83. https://doi.org/10.2307/3001968

Code

Analysis code at https://github.com/gildedpterodactyl/BRSM-Project

Contributions

Three of us have contributed approximately equally. Each of us have done sections of code and analysis for the following as mentioned below:

Ananya Kasavajhala: Descriptive Analysis, Theory and Interpretation of performance patterns across conditions, report integration

Rohit Jeswanth: Data Preprocessing, Feature Extraction and Inferential Statistical Analysis including correlation, ANOVA and t-tests, code and repository integration and maintenance

Vishak Kashyap K: Level-wise Analysis, Reliability Analysis, supporting visualizations and verification of statistical outputs, presentation slides integration

***